

Mahesh Patel.

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EDUCATION

Thadomal Shahani Engineering College Mumbai University <i>B.E. in Artificial Intelligence and Data Science (CGPA: 8.23)</i>	<i>Aug 2023 – June 2027</i> <i>Mumbai</i>
SVKM's Mithibai Junior College <i>Higher Secondary Certificate (HSC: 88.5%)</i>	<i>Aug 2021 – March 2023</i> <i>Mumbai</i>

EXPERIENCE

Artificial Intelligence Intern Hiranandani Group, Mumbai – Built CardVault , a PWA application to digitize and manage 1,000+ physical business cards, leveraging Google Gemini Vision API to extract contact details, company info, and service catalogs from card photos into MySQL. – Implemented a smart search engine combining MySQL full-text search with AI query expansion, caching, and role-based access control. – Contributed to the development of a real-time Gate Check-In/Check-Out application for site personnel tracking and the enterprise Purchase Order website.	<i>June 2026 – July 2026</i>
AI/ML Research Intern Thadomal Shahani Engineering College, Mumbai – Graph Neural Networks: Conducted a comparative study of GCNs, GATs, and custom hybrid models for node classification, validating that optimized/fine-tuned GCNs provide a more viable and efficient solution than hybrid architectures. – Reinforcement Learning via Verifiable Rewards (RLVR): Developed a step-level gradient routing framework (EA-GRPO / KL-GRPO) addressing credit assignment in GRPO by selectively masking mastered steps; achieved 72.94% on MATH-500 (+1.04% vs. standard GRPO) and 15.83% on AIME 2025 (+2.83%) on Qwen3-4B with zero wall-clock overhead.	<i>Dec 2025 – Jan 2026</i>

PROJECTS

Municipal Grievance Routing Engine Python, TensorFlow, FAISS, RAG, Scikit-Learn  GitHub – Built a multilingual NLP pipeline for automated grievance classification and department routing across Hindi, English, and Hinglish using a TextCNN + BiLSTM + Attention architecture with a custom Byte-Pair Encoding tokenizer. – Integrated a FAISS-based RAG engine for contextual FAQ retrieval, enabling dynamic response generation without retraining.	
Warden Security Log Analytics & Threat Detection Python, SQLite, Linux, HTML5  GitHub – Built a security log parser in Python using custom regex to extract login and privilege escalation events from Linux system logs into SQLite. – Created detection rules for SSH brute-force, botnet attacks, and compromised credentials, reducing log analysis time via automated HTML audit reports.	

TECHNICAL SKILLS

Languages: Python, C++, C, SQL
AI, ML & NLP: TensorFlow, PyTorch, Scikit-Learn, LangChain, FAISS, RAG, NumPy, Pandas
Web & Backend: FastAPI, Flask, REST APIs, Progressive Web Apps (PWA)
Databases: PostgreSQL, MySQL, SQLite, Supabase
Tools & Analytics: Linux, Git, GitHub, Docker, Postman, AWS, Power BI, Tableau, Excel

POSITIONS OF RESPONSIBILITY

Envoy and Placement Coordinator Training and Placement Cell - TSEC – Coordinated pre-placement talks and hiring protocols with companies while managing recruitment databases. – Organized campus interviews and guided students on resumes, applications, and placement protocols.	<i>July 2026 – Present</i>
Design Head TSEC CodeTantra – Led the design team for the flagship annual hackathon, coordinated cross-functional deliverables under tight deadlines, and contributed to the annual department magazine.	<i>Mar 2025 – Mar 2026</i>

ACHIEVEMENTS

Augenblick Hackathon 2026, Finalist	<i>March 2026</i>
IES MCRC Hackathon 2.0, 2nd Rank in Round 1 and Round 2 Qualifier	<i>March 2025</i>